

End-semester Examinations Winter Semester 2024-25
Course: Understanding Human Behaviour
Part -C (Unit-3)

Total Marks = 12.5

Questions:

1. a) Differentiate between neuron cells and glial cells. *(1 mark)*

b) Name the two divisions of the autonomic nervous system and explain. *(2 marks)*
2. a) Amit refuses to swim in the ocean because he recently saw a viral video of a shark attack, even though such incidents are extremely rare. Identify the heuristic and explain. *(0.5 + 1 marks)*

b) Explain functional fixedness *(1 mark)*
3. What do you mean by “hemispheric specialisation”? *(1 mark)*
4. Limbic system regions help guide control of the internal environment, and the hypothalamus is one such region. Name and mention the main function of the three output pathways of the Hypothalamus to affect changes in the internal state. *(3 marks)*
5. Write a short note on the parts (at least 3 parts) of the brainstem and their functions. *(3 marks)*

Answers:

1. a) **Neurons** are the basic signalling units that transmit information throughout the nervous system. *(0.5)*

Glial cells serve various functions in the nervous system, providing structural support and electrical insulation to neurons, modulating neuronal activity, cleaning up cellular debris, and forming the blood-brain barrier. *(0.5)*

b) The two divisions of the autonomic nervous system are:

Sympathetic Nervous System – It prepares the body for "fight or flight" responses during stressful or emergency situations. It increases heart rate, dilates pupils, and redirects blood flow to muscles. *(1)*

Parasympathetic Nervous System – It promotes "rest and digest" activities. It slows the heart rate, stimulates digestion, and helps conserve energy. *(1)*
2. a) **Availability heuristic (0.5)**. It is a mental shortcut where people judge the likelihood of an event based on how easily examples come to mind. *(1)*
b) It is a tendency to think of using objects only in ways they have been used before *(1)*
3. “**Hemispheric Specialisation**” refers to the tendency for specific mental processes and behaviours to be more dominant (or specific) in one hemisphere of the brain than the other *(1)*.
4. Three output pathways of the Hypothalamus are:

- a) *Autonomic* = stimulates sympathetic and parasympathetic pathways (*1*)
- b) *Neuroendocrine* = regulates hormones (*1*)
- c) *Motivational* = putting together complex, flexible, nuanced behaviours (*1*)

5. The parts of the brainstem are: (*0.5+0.5 each, any 3*)

- a) Medulla: controls vital functions such as respiration, heart rate, and arousal
- b) Pons: Arousal and pain, REM sleep, connection between cerebral cortex and cerebellum
- c) Cerebellum: Regulation of motor activities, forward model, balance and coordination
- d) Superior colliculus: visual perception of peripheral objects, eye reflexes, and sharpening the view
- e) Inferior colliculus: locating and orienting towards an auditory stimulus
- f) Red nucleus: motor coordination in children, shoulder movement